



Speech transcription from South Tyrolean Dialect to Standard German with Whisper

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Abstract

This study presents the first fine-tuned *Whisper* model for the automatic translation of South Tyrolean dialectal speech into Standard German text. To address an unmet need for subtitling and translation, we introduce a small corpus of manually annotated and synthetic speech data compiled for this task. Through fine-tuning and hyperparameter optimisation, our model achieves a BLEU score of 86.18 significantly outperforming baseline error rates. Our findings highlight *Whisper*'s effectiveness in handling dialectal speech, contributing to low-resource language research. The model is already being used in a heritage collaboration for large-scale translation of audiovisual archival material and is also being considered for application in news broadcasting and tourism promotion. Future directions include expanding the training data and extending hyperparameter optimisation to improve the model's performance and generalisation across South Tyrolean dialectal variations.

Index Terms: speech-to-text, translation, large language models, *Whisper*, low-resource languages, dialects, German

1. Introduction

The South Tyrolean dialect (hereafter South Tyrolean), spoken in Italy's northernmost province of South Tyrol (Südtirol), is a protected Austro-Bavarian variety blending Germanic and Romance linguistic elements (Italian and Ladin) [1]. Unlike South Tyrolean German, the region's standard German variety, South Tyrolean is primarily oral and lacks a standardised orthography. Widely used informally across generations, South Tyrolean is favoured for casual communication like text messages and social media, while formal contexts rely on the standard variety.¹ Although unique, South Tyrol's sociolinguistic situation parallels many European regions where regional dialects dominate informal oral communication within diglossic or diglossic settings [3].

Limited subtitling of dialectal speech in formal media contexts restricts access for individuals with hearing impairments or those unfamiliar with South Tyrolean, such as visitors or non-native residents. This study responds to an explicit demand from local community beneficiaries for an automated solution to transcribe and subtitle South Tyrolean speech into Standard German, introducing the first *Whisper* fine-tuned model for this purpose.

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¹According to [2, p. 1], 98% of German-speaking South Tyroleans report being able to speak the dialect fluently.

2. Related work

In recent years, strides have been made in the field of NLP for Germanic dialects, enhancing the accessibility and usability of these linguistic varieties in digital formats [4, 5, 6, 7, 8]. With regard to spoken data, [9] lists 39 audio datasets covering low and high German dialects. The *AnDy* project uses AI and the *REDE* corpus [10] to automatically analyse the dynamics of German dialectal speech, while a 343-hour corpus of read Swiss German speech annotated with Standard German text has been used to fine-tune an XLS-R translation model [11]. Swiss German and Luxembourgish are the only Germanic dialects to have been processed for translation into Standard German. Swiss German studies include hybrid Hidden Markov Models [12], Time Delay Neural Networks with *Kaldi* [13] and pronunciation mapping to Standard German [14], as well as transfer learning with *Jasper* and *DeepSpeech* [15, 16] and on other well-performing hybrid systems [17]. Luxembourgish research fine-tuned *Whisper* and *wav2vec 2.0* on 67 hours of annotated speech, achieving better results with *Whisper* [18].

Building on the speaker-centric NLP perspective advocated in recent discussions on low-resource language colonising discourse [19], a survey of 327 speakers of German varieties found strong support for systems transcribing dialect to Standard German (61%) [6]. While this result may not be fully representative of the broader population, they highlight interest in advancing dialect NLP and spoken language processing technologies.

3. Method

3.1. Datasets

3.1.1. GOLD

As is often the case with under-resourced languages, no adequate corpora exist yet for the development of a Speech-to-Text/Translation system for South Tyrolean². For this reason, we are creating a corpus of labelled South Tyrolean-to-Standard German data by combining different publicly available sources, both openly licensed and proprietary, with our own speech recordings. Publicly available sources span spontaneous and scripted speech, include both proprietary and openly licensed language datasets, and capture a wide range of dialectal diversity. The dataset currently comprises 412 minutes (6 hours and 52 minutes) of speech drawn from four distinct spoken sources (ARB, ALP, EUR, IDM), down-sampled to 16kHz, totalling approximately 4,000 sentences or data points, as summarised in Table

²Multilingual speech datasets consulted: *Mozilla CommonVoice*, *Google Cloud Speech-to-Text V2*, *VoxLingua 107*, *Facebook Vox Populi*, *VoxForge*, *Multilingual LibriSpeech Corpus*, *M-AIABS Speech Dataset*, *VoxCeleb*, *CMU Wilderness Multilingual Speech Dataset* and *FLEURS*.

Table 1: Overview of the GOLD corpus, detailing speech type, duration, average sentence length in characters, the number of tokens, types and token/type ratio of the Standard German reference text, number of unique speakers and estimated age group of the speakers.

Dataset	Speech type	Hours	avg(L)	Tokens	Types	TTR	Speakers	Age
ARB	read	00:47:56	44.47	5,533	1,233	0.222	3	25 – 45
ALP	read, spontaneous	04:47:23	42.31	36,144	146	0.004	180	15 – 86
EUR	read, spontaneous	00:13:22	115.40	1,508	457	0.303	3	25 – 45
IDM	spontaneous	01:03:53	120.85	8,599	2,472	0.287	86	25 – 75
TOTAL		06:52:34		51,784			273	

1. We must partly rely on proprietary data because openly licensed datasets for this task are scarce.

ARB. Scripted speech from the publicly available *DaZUgeHÖREN* textbook for learners of South Tyrolean. The textbook includes CD tracks featuring exercise prompts spoken in dialect, accompanied by printed translations in Standard German. Data acquisition involved converting the MP3 tracks into WAV format, manually transcribing the corresponding printed translations and manually aligning audio and text with ELAN³. As per *Whisper* requirements, audio files exceeding 30 seconds were manually chunked into shorter segments. The correspondence between audio and text in this dataset is not always accurate, with occasional deviations in the printed text.

ALP. Formal and informal speech from the crowdsourced multilingual speech corpus *AlpiLinK* (v1.1.3) [20]. We extracted the South Tyrolean speech data collected through the Standard German translation task, which amounts to a total 30 sentences uttered by multiple speakers.

EUR. Our own speech recordings of human translators spontaneously producing spoken dialect translations of Standard German sentences drawn from newspaper texts provided with *Korpus Südtirol*⁴. The participants contributing to this speech corpus—which will be made freely available to the scientific community—are affiliated with our home institution, and were required to sign a form consenting to the use and the open access publication of their voices. They are shown Standard German sentences, selected in a manner similar to that of [21, p. 2], asked to translate them to South Tyrolean speech, and given the option to revise their translation. For each participant, we collect information about their age, gender, provenance and dialectal variant spoken. Speech is recorded in segments of under 30 seconds using a dynamic cardioid vocal microphone⁵. Each segment is then manually aligned with its corresponding reference text, where the spoken South Tyrolean translation serves as a direct, literal representation of the Standard German text.

IDM. Spontaneous speech from 44 subtitled videos posted on YouTube by IDM Südtirol⁶. The videos are promotional in nature and cover a wide range of topics. We downloaded the videos as MP4 files through the Python pytube library⁷. In a second step, we split the videos into frames, increased the contrast of the frames to make the text stand out more prominently against the background and ran *PaddleOCR* [22] on each frame cropping to the bottom 1/3 of the image. Errors in the resulting OCR were mainly caused by misrecognised special characters (ß, ä, ü, ö), video quality and textual background interference. The OCR was then manually corrected so as to obtain a

clean reference text for each video. Next, we clipped the audio—which often contains background music—in WAV format from the videos, manually aligning audio segments of under 30 seconds to the corrected OCR using ELAN. The dataset containing the audio and text information was obtained thanks to a custom Python script used to process ELAN EAF files. As these are subtitled videos, the correspondence between audio and text in this dataset is rarely literal.

3.1.2. SYNTH

The synthetic dataset is derived from the first 119 episodes of the *Budl&Stubn* podcast⁸. The episodes were downloaded as MP3 files from *SoundCloud* using the Python *soundcloud-lib* library⁹. The dataset totals 7,127.35 minutes (118 hours and 47 minutes) of speech. The translated transcription was performed using the *Whisper* large-v3 model, with diarisation enabled through the Python *pyannote* library. After filtering out sentences shorter than 2 seconds, the final synthetic training dataset comprised 6,729 chunks.

3.2. Model

We chose to work with *Whisper* (large-v3 model) over its prominent contender, *wav2vec 2.0*, primarily due to findings from [18], whose study closely aligns with our objectives. Their research demonstrated that *Whisper*’s semi-supervised architecture surpasses *wav2vec 2.0* in handling poor audio quality and multilingual speech, and in output quality, particularly in terms of capitalisation and punctuation. This is a critical factor for Germanic languages where capitalisation rules are stringent. However, as [18] do not address translation tasks, we also turned to [15] for insights on adapting *Whisper* for translation.

Each dataset was split 80-20, with 80% allocated for training and 20% for testing. Within the training set, 20% was used for validation. We selected the Word Error Rate (WER) as evaluation metric for automatic speech recognition and BLEU for translation. Confidence intervals (CI) were obtained with bootstrap sampling with 1000 samples. All of the experiments were conducted on the high-performance computing infrastructure of the South Tyrol Scientific Network¹⁰, using an AMD EPYC 7543P 32-Core Processor (CPU) and an NVIDIA A100 Ampere 40GB Graphics Card (GPU).

4. Results and discussion

4.1. Baseline

A baseline analysis of the *Whisper* large-v3 model (1.5B parameters) set to German and performed on the GOLD dataset re-

³Version 6.4. <https://archive.mpi.nl/tla/elan>

⁴<http://www.korpus-suedtirol.it/index.html>

⁵Behringer ULTRAVOICE XM8500.

⁶<https://www.youtube.com/@suedtirol.official>

⁷<https://pytube.io/en/latest/index.html>

⁸<https://soundcloud.com/budlundstubn>

⁹<https://pypi.org/project/soundcloud-lib/>

¹⁰<https://www.unibz.it/en/services/ict-services/>

Table 2: Hyperparameter configurations for gold, optGold and optSynth.

Hyperparameters	gold	optGold		optSynth	
		Range	Best	Range	Best
adam_beta1	—	0.85, 0.99	0.965	0.872, 0.947	0.899
adam_beta2	—	0.95, 0.999	0.959	0.966, 0.996	0.972
adam_epsilon	1×10^{-8}	1×10^{-9} , 1×10^{-6}	1.44×10^{-8}	5.71×10^{-9} , 1.03×10^{-7}	2.09×10^{-8}
batch_size	4	4, 8, 16	4	4, 8	4
dropout	—	0.0, 0.2	0.090	0.039, 0.093	0.074
grad_accum_steps	1	1, 2, 4	1	2, 4	4
learning_rate	5×10^{-6}	1×10^{-8} , 1×10^{-4}	1.50×10^{-6}	8.02×10^{-8} , 7.69×10^{-6}	2.42×10^{-7}
lr_scheduler	—	linear, cosine, constant	linear	linear, cosine, constant	linear
max_grad_norm	—	0.1, 5.0	3.699	1.140, 3.460	2.729
num_train_epochs	10	5, 20	18	6, 13	8
warmup_steps	100	50, 200	177	87, 192	129
weight_decay	0.01	1×10^{-6} , 1×10^{-2}	3.19×10^{-5}	1.38×10^{-6} , 7.73×10^{-3}	8.51×10^{-6}

sulted in a WER of 0.53 and a BLEU score of 33.95, indicating a moderate level of accuracy with notable errors, while still demonstrating a good degree of similarity between the predicted and the reference translation.

4.2. Preliminary fine-tuning

In an initial experiment we fine-tuned a model on the GOLD dataset, explicitly setting the language argument to German¹¹ and using the generic hyperparameter configuration listed in Table 2 (column gold). As shown in Table 3, the model demonstrated a significant reduction in WER and improvement in BLEU compared to the baseline. Increasing the number of training epochs to 50 did not improve performance; nevertheless, to assess the impact of this parameter configuration on boosting *Whisper*’s performance on our small datasets, we conducted hyperparameter optimisation.

4.3. Optimisation

To gauge the best hyperparameter configuration for model fine-tuning, we conducted multiple optimisation experiments on GOLD (hereafter optGold) and on SYNTH (hereafter optSynth) with the *Optuna* framework [23]¹². Our choice of hyperparameter ranges was informed by existing literature, but we extended these to explore a broader spectrum of possibilities.

optGold20. Starting from the range shown in Table 2 and setting *Optuna*’s optimisation multivariate parameter to false to examine the impact of *Whisper*’s hyperparameters in isolation [24], this experiment ran 20 training trials in 20 hours. The best configuration achieved a 12% WER and a BLEU score of 78.90 on GOLD, improving on the preliminary fine-tuning experiment across all error metrics. These results were influenced by five key parameters, which together accounted for most of the model’s performance variance. In order of importance, Adam ϵ had the greatest impact, followed by number of epochs, warm-up steps, learning rate and Adam β . The dominance of Adam ϵ suggests that its small 1.44×10^{-8} value (the same as recommended by [25]) was crucial for stable and effective learning while a higher value could have caused larger fluctuations in gradients, leading to poor convergence. Training duration and learning rate scheduling were closer to the higher end of the value range we set, suggesting that longer training and more aggressive learning rate scheduling helped achieve

optimal performance. Finally, the Adam β_1 value (0.96) facilitated smoother updates while slowing adaptation to new patterns (convergence delay). Similarly, the β_2 value (0.95) would indicate a more stable but slower adjustment of the learning rate.

optGold70. To further investigate the impact of hyperparameter optimisation, we conducted a second experiment with *Optuna*’s multivariate parameter set to true and 70 trials (Figure 1). The process ran for 41:08:57 hours, allowing for a more comprehensive exploration of parameter interactions. However, while the results showed no performance improvement over optGold20, indicating that a wider search space did not yield beneficial parameter configurations, weight decay and learning rate showed a higher level of importance.

optSynth20. For the optSynth study, we ran an experiment with *Optuna*’s multivariate parameter disabled and another with it enabled. For the former experiment, we increased the GPU and RAM size, as well as the trial numbers to 20, for a total training runtime of 920 minutes (15:20:38 hours). The best trial from this study, presented in Table 2, used SGD over AdamW as the optimiser. While slightly better than the baseline, its overall performance on the test set remained poor. To further investigate this result, we retrained the best checkpoint on SYNTH for 50 epochs. However, the performance showed only negligible improvements compared to training the same model for 8 epochs and the baseline, suggesting that increasing the number of epochs had no meaningful impact. Indeed, the most influential hyperparameters for this best trial, in order of importance, were: Adam β_2 (0.97), dropout rate (7%), max gradient norm (2.72), learning rate (very small at 2.42×10^{-7}) and warm-up steps (129). In the latter experiment, the model’s optimisation progress after 13 epochs showed no improvement on the error rates of optSynth20, leading us to conclude that that *Optuna*’s multivariate parameter had little to no bearing on the convergence behavior for this particular benchmark.

4.3.1. optGold20 output

Below, we present a short selection of the model’s predictions (Hyp) alongside the reference translations (Ref), focussing on cases with the highest error rates. *Whisper*’s errors can be grouped into several categories. A common issue is hallucinations, with the model generating words not present in the audio or duplicating utterances. It also struggles to recognise some named entities, often replacing them with more common (Example 1), phonetically similar or even invented names. Occasionally, it misrecognises gender, transforming female-referenced sentences into male ones (Example 2). Another is lexical reformulation, where the model substitutes words with

¹¹Applied to both *Whisper*’s tokeniser and decoder. Omitting the language specification resulted in markedly poor results, which are not worth reporting.

¹²<https://optuna.org/>

Table 3: Performance of Whisper’s baseline and fine-tuned large-v3 models on GOLD. Confidence intervals (CI) were obtained with bootstrap sampling over 1000 samples.

Model	Epochs	WER	WER CI	BLEU	BLEU CI	Resources	Train time
baseline	—	0.53	0.49, 0.57	33.95	30.68, 37.23	—	—
gold	10	0.19	0.15, 0.23	78.90	74.95, 82.59	CPU:1, A100(3G,24GB):1, GPU:1, Mem:24G, Node:1	09:04:11
gold	50	0.28	0.25, 0.32	61.75	58.56, 65.00	CPU:1, A100(3G,24GB):1, GPU:1, Mem:24G, Node:1	—
Study	Epochs	WER	WER CI	BLEU	BLEU CI	Resources	Optim. time
optGold20	6	0.12	0.09, 0.16	86.18	82.07, 89.55	CPU:1, A100(3G,40GB):1, GPU:1, Mem:32G, Node:1	01:03:14
optGold70	14	0.15	0.12, 0.18	82.60	78.70, 86.13	CPU:1, A100(3G,80GB):1, GPU:1, Mem:64G, Node:1	00:28:19
optSynth20	8	0.55	0.50, 0.59	33.67	30.94, 36.26	CPU:1, A100(3G,80GB):1, GPU:1, Mem:64G, Node:1	00:46:58
optSynth20	50	0.54	0.50, 0.59	33.79	30.76, 36.31	CPU:1, A100(3G,80GB):1, GPU:1, Mem:64G, Node:1	03:27:43

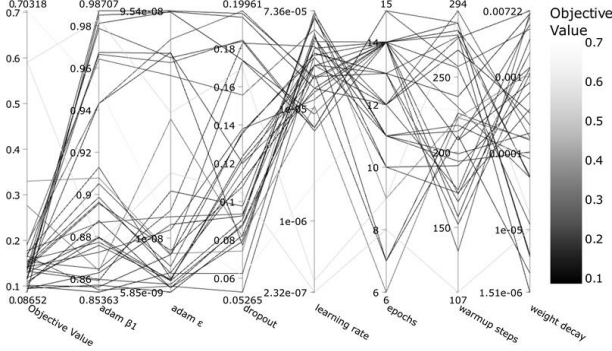


Figure 1: Parallel coordinate plot of optGold70 showing optimisation progress across 70 trials, with darker lines representing the best performing trials.

synonyms, altering the phrasing while preserving meaning (Example 3). Some errors stem from data preprocessing, where insufficient padding at the start of audio chunks causes the first word to be misinterpreted. In other cases, the model accurately translates the audio, whereas the reference text presents a reformulation.

- Dataset:** ALP; **WER:** 0.14; **BLEU:** 84.08
Ref: Sara sagt, dass wir zu viel essen. — **Hyp:** Lisa sagt, dass wir zu viel essen.
- Dataset:** ALP; **WER:** 0.25; **BLEU:** 53.03
Ref: Martina war draußen unterwegs, hast du sie gesehen? Was hat sie gemacht? — **Hyp:** Martin war draußen unterwegs, hast du ihn gesehen? Was hat er gemacht?
- Dataset:** EUR; **WER:** 0.43; **BLEU:** 29.15
Ref: Die Nachbarprovinz Trient musste da mit fünf Prozent weit höhere Werte verzeichnen, ganz abgesehen vom restlichen Staatsgebiet, das durchschnittlich rund zwölf Prozent aufweist. — **Hyp:** Die Nachbarprovinz Trient hat damit 5 Prozent weit höhere Werte verzeichnen müssen, ganz angesehen vom restlichen Staatsgebiet, das durchschnittlich um 12 Prozent aufweist.
- Dataset:** IDM; **WER:** 0.30; **BLEU:** 56.59
Ref: In meinem Umfeld haben die Natur und der Wein die wichtigste Rolle gespielt. — **Hyp:** wo um mich herum die Natur und der Wein die wichtigste Rolle gespielt haben.

4.4. Limitations

The fine-tuning dataset used is limited in size and lacks balance among the different South Tyrolean variations, which may affect the generalisability of the findings and introduce biases due to the underrepresentation of certain dialects.

In compliance with submission anonymisation rules, access to the code and the reusable portion of the training datasets will be provided upon acceptance of this submission.

4.5. Ethical considerations

Participants of the EUR corpus consented to the use of their anonymised contributions and their involvement was entirely voluntary. No metadata or personal identifiable information was used in the training or evaluation of the model.

5. Conclusion and future work

In this study, we fine-tuned a Speech-to-Text model for translating South Tyrolean dialect into Standard German using OpenAI’s *Whisper*, concurrently compiling a first corpus for this purpose. Compared to the baseline, our optimised model reduced the WER from 53% to 12% and achieved a BLEU score of 86.18, reflecting a higher-quality transcription. The best performance was obtained using targeted hyperparameter optimisation, particularly for AdamW parameters, learning and dropout rates. While the model performed well on manually annotated data, results on the synthetic dataset remained suboptimal, indicating a need for further refinement. Despite dataset size and resource limitations, our approach successfully improved dialectal speech processing, showing promise for practical applications in subtitling and archival documentation. Indeed, our model is currently deployed in partnership with a local heritage beneficiary, the Office for Film and Media of the Autonomous province of Bolzano, to mass-translate audiovisual archives. The Office will review and correct the translations, providing feedback to refine the model and ensuring its practical relevance to community-specific needs. The model is also being considered for deployment in news broadcasting and tourism promotion.

Future work will focus on expanding and combining training data, extending optimisation experiments to include a much larger number of trials and exploring alternative multitask models such as *Seamless M4Tv2*¹³, *distil-whisper-large-v3*¹⁴ and *NVIDIA Canary*¹⁵ for comparative analysis and improved generalisation.

¹³https://huggingface.co/docs/transformers/en/model_doc/seamless_m4t_v2

¹⁴<https://huggingface.co/distil-whisper/distil-large-v3>

¹⁵<https://huggingface.co/nvidia/canary-1b>

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